

Features: AI (2pts), non-AI (1pts)

AI Features

- **Train AI bot with your preference method:** *A**, *reinforcement-learning*, *BFS*, etc.
 - **A*:** chase the prisoner or run to the tunnel. Hard uses minimax with alpha-beta pruning, looking 4 to 6 moves ahead. This also lets you play solo if no second computer joins.
 - **Reinforcement-learning:** Q-learning bot. The 5×5 map is small enough to train a *reinforcement-learning* bot. Show a training graph in the demo of the bot going from random to smart.
- ***Live win-probability bar with pop-up meme.** The server runs 200 quick random simulations from the current position. If the prisoner wins 130 of them, the bar shows 65% for the prisoner. It shifts every turn, which looks great in a demo.
 - **Check your moves.** After each move, the coach asks: "Did this make your chances of winning better or worse?" and come up with the related meme + sound effects!

Non AI Features

Game mode: 1) PvP 2) stage (AI difficulty + map size) 3) special mode

For every game mode:

- ***Map balancer*.** initially have random (fair) 100 maps and randomly pick one before each match, so neither role has an unfair map. BFS still checks that there is no deadlock. Don't let la policia get too close to the tunnels.
- **Power-ups (picked up on the map or limited per match)**
 - Prisoner: **Dig** (remove one obstacle), **Sprint** (move 2 blocks), **Smoke bomb** (the warder can't see you for 1 turn).
 - Warder: **Radar** (reveal the prisoner), **Roadblock** (place one temporary obstacle)

2) Stage (each map per stage)

- **Pick an AI opponent with 3 difficulty levels.** Easy moves randomly. Medium uses BFS or A* to chase the prisoner or run to the tunnel. Hard uses minimax with alpha-beta pruning, looking 4 to 6 moves ahead. This also lets you play solo if no second computer joins.
- **Selectable map size (5×5, 7×7, 9×9).** Keep the 20% obstacle ratio, so 7×7 gives about 10 obstacles.

3) Special mode (5 mode)

- **Obstacle shifting.** On your turn you either move your character or push one obstacle by one block. The server must rerun BFS to reject any shift that causes a deadlock.

- **Key and tunnel.** The prisoner must pick up a key somewhere on the map before the tunnel opens.
- **Fake tunnels.** The map has 3 tunnel blocks. Only the prisoner knows which one is real, and the warder has to guess.
- **Shrinking prison.** Every 3 rounds a random free block collapses into an obstacle. This guarantees the match ends and adds pressure.
- **Shrinking timer.** The turn timer drops from 10s to 5s as the match goes on.